05 - System Architecture and Technology Stack

Docker services, data flow, technologies

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

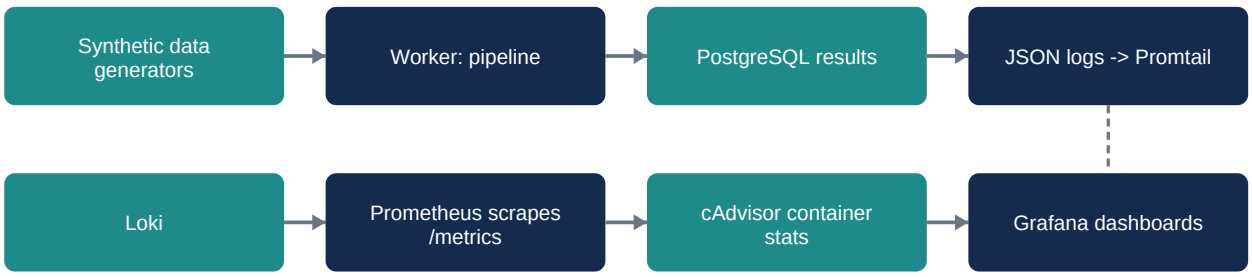
Source files used for this document

docker-compose.yml; documentation/system_architecture.md; documentation/technology_stack.md; requirements.txt; requirements-dev.txt

Services in docker-compose.yml

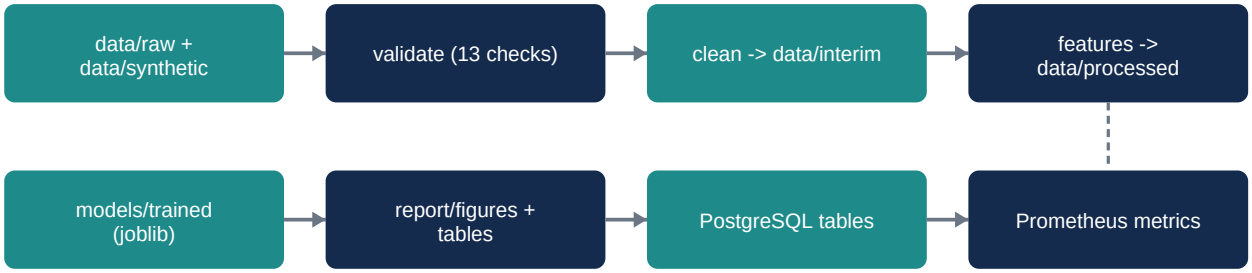
Service	Image	Ports	Depends on	Healthcheck
postgres	postgres:16-alpine	POSTGRES_PORT:-5432:5432	-	yes
worker	built from infrastructure/docker/Dockerfile.worker	EXPORTER_PORT:-8000:8000	postgres	yes
jupyter	built from infrastructure/docker/Dockerfile.jupyter	JUPYTER_PORT:-8888:8888	postgres, worker	yes
prometheus	prom/prometheus:v2.53.1	PROMETHEUS_PORT:-9090:9090	-	yes
loki	grafana/loki:3.1.1	LOKI_PORT:-3100:3100	-	yes
promtail	grafana/promtail:3.1.1	-	loki	no
cadvisor	gcr.io/cadvisor/cadvisor:v0.49.1	CADVISOR_PORT:-8081:8080	-	yes
grafana	grafana/grafana:11.2.0	GRAFANA_PORT:-3000:3000	prometheus, loki, postgres	yes

Architecture diagram



Worker writes results to PostgreSQL and logs to a shared volume; Promtail ships logs to Loki; Prometheus scrapes worker, cAdvisor, Loki and Grafana; Grafana reads Prometheus, Loki and PostgreSQL.

Data flow



Where each artefact is written by src/pipeline.py.

Start-up order enforced by health checks

postgres (pg_isready) -> worker (healthy only when the metric fintech_pipeline_ready equals 1) -> jupyter. Grafana waits for prometheus, loki and postgres; promtail waits for loki.

Technology stack

Layer	Technology	Reason
Language	Python 3.11 in containers	Data-science ecosystem
Data	pandas, NumPy, SciPy, PyArrow	Tables, statistics, Parquet
ML	scikit-learn, XGBoost, joblib	Models, tuning, serialization
Plots	Matplotlib, Seaborn, Plotly	Static and interactive figures
Database	PostgreSQL 16 via SQLAlchemy + psycopg2 (SQLite fallback)	Results and metadata
Metrics	prometheus-client, Prometheus v2.53.1	Pull-based time series
Logs	JSON logging, Promtail, Loki 3.1.1	Searchable logs
Dashboards	Grafana 11.2.0 provisioning	Dashboards and alerts as code
Containers	Docker Compose, cAdvisor v0.49.1	One-command environment, container stats
Research UI	JupyterLab	Interactive notebooks
Quality	pytest, Makefile	Tests and shortcuts

Dependencies declared

```
# Core runtime dependencies (worker + notebooks). Ranges are tested together on Python 3.11-3.14.
# Install:  pip install -r requirements.txt     (notebooks/tests: also requirements-dev.txt)

# --- data & numerics -----------------------------------------------------
numpy>=1.26,<3            # array maths
pandas>=2.2,<4            # dataframes
scipy>=1.11,<2            # statistical tests (Mann-Whitney, chi-square, Spearman, ...)
pyarrow>=15               # Parquet storage

# --- machine learning ----------------------------------------------------
scikit-learn>=1.4,<2      # LR, Random Forest, Isolation Forest, LOF, CV/tuning, metrics
xgboost-cpu>=2.1,<4; sys_platform == "linux"   # gradient-boosted trees (CPU build: avoids ~350 MB of NVIDIA G
xgboost>=2.0,<4; sys_platform != "linux"         # same library on Windows/macOS
joblib>=1.3               # model serialisation

# --- visualisation -------------------------------------------------------
matplotlib>=3.8,<4
seaborn>=0.13,<1
plotly>=5.20,<7
tabulate>=0.9             # DataFrame.to_markdown for report tables

# --- infrastructure ------------------------------------------------------
prometheus-client>=0.20   # /metrics exporter
SQLAlchemy>=2.0,<3        # database access
```

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psycopg2-binary>=2.9      # PostgreSQL driver
PyYAML>=6.0               # configuration

-r requirements.txt

# --- interactive research & automated checks -------------------------------------
jupyterlab>=4.1,<5
ipykernel>=6.29
nbformat>=5.10
nbclient>=0.10
nbconvert>=7.16
pytest>=8.0
python-pptx>=1.0,<2   # review decks (review/scripts)
```

Not used by design: React/HTML front end, SMOTE oversampling, deep learning, Alertmanager (listed as lab-scope omissions in the technology document).